Special Topics: Social Media Mining and Analysis – Assignment 2

Multimodal Hateful Meme Classification

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https://github.com/BogdanFanea/SpecialTopicSocialMining

We adapted Fusion and Orthogonal Projection (FOP) [1, 2] to classify memes using both the image and its supplied text. We reuse frozen embeddings from Assignment 1 and compare weighted addition, gated fusion, and gated fusion with an orthogonal projection loss.

1. Data and evaluation labels

Split	Examples	Non-hateful	Hateful
Training	8,500	5,450	3,050
Validation (dev)	500	250	250
Held-out test	1,000	—	—

We kept the course Hateful Memes splits [3] and their supplied training and validation labels. The splits have no shared IDs. For each split, we checked the cached feature dimensions, annotation hashes and exact ID membership, then joined image and text embeddings by meme ID. The course test file has no labels. We report validation AUROC and export one held-out probability per ID.

2. Encoders and preprocessing

Input	Exact pretrained checkpoint	Dim.	Pooling
Image	`openai/clip-vit-base-patch32`	512	Vision CLS + projection
Text	`sentence-transformers/all-mpnet-base-v2`	768	Masked mean + L2

CLIP and MPNet had the highest mean validation AUROC in their respective modalities in Assignment 1. Each fusion model uses the same frozen pair. CLIP [4] takes RGB pixels after bicubic resizing, a 224 × 224 center crop and checkpoint channel normalization. MPNet [5] uses the supplied transcript, its matching tokenizer, dynamic padding and a 384-token limit. Masked pooling excludes padding, then the embedding is L2-normalized. Exact revisions are recorded in the checkpoint metadata and Assignment 1 configuration.

3. Fusion, training and model selection

Each projection maps its input to 128 dimensions with Linear, BatchNorm, ReLU, dropout ($p = 0.5$), then rowwise L2 normalization. Following FOP [1, 2], the gated model concatenates the projected vectors u and v. A Linear(256,128), BatchNorm, ReLU and Linear(128,128) network followed by sigmoid produces gate g. Fusion combines tanh(u) and tanh(v) elementwise with weights g and 1 minus g. The linear addition baseline uses two learned scalar weights on u and v. A Linear(128,2) head predicts raw logits. Softmax class 1 is the hateful probability at inference.

All variants minimize cross-entropy on integer labels. For CE+OP we add a weighted penalty on L2-normalized fused vectors. It is one minus the mean same-class cosine similarity, plus 0.7 times the mean absolute cross-class cosine similarity. Same-class self-pairs are excluded and empty means contribute zero. The pairs are whole memes, with each original image still paired to its transcript.

We trained the projection, fusion and head in NumPy on CPU using AdamW (learning rate 0.001, weight decay 0.01), 20 epochs, and seeds 42, 43 and 44. For each epoch, we independently shuffled each class, split it into 67 chunks, paired the chunks, and shuffled within batches. This uses every example once and includes both classes at roughly their original proportions. Batches contain at most 128 examples.

For each run we retained the earliest epoch with maximum validation AUROC. Positive OP weights 0.1, 0.3 and 1.0 were compared by mean validation AUROC. The zero-weight gated ablation uses the same settings, initialization, batch order and dropout stream per seed. Official held-out labels are unavailable. After selection, we exported test probabilities for the highest-mean configuration using the predetermined seed 42.

Selected epochs for seeds 42/43/44. Weighted addition: 9/15/19. Gated CE: 13/11/14. Gated CE+OP: 13/11/14.

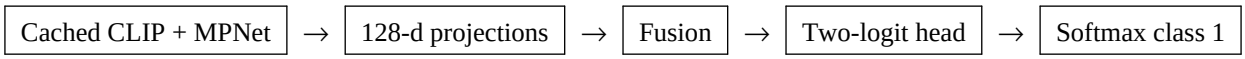

Figure 1: Multimodal classification pipeline.

4. Results and comparison

AUROC uses continuous hateful-class scores. Values are validation means and sample standard deviations across seeds 42, 43 and 44.

Table 1: Validation AUROC. A1 marks reused baselines.

Model	Mean AUROC	SD	Trainable
CLIP image only (A1)	0.6445	0.0049	513
MPNet text only (A1)	0.6390	0.0031	769
Weighted addition + CE	0.6806	0.0086	164,868
Gated fusion + CE	**0.7281**	**0.0031**	**214,530**
Gated fusion + CE + OP (0.1)	0.7256	0.0024	214,530

Gated CE improved over CLIP by **+0.0836** AUROC and over weighted addition by **+0.0475**. Gating beat addition in all three seeds, though it also adds capacity. The best positive OP weight scored **-0.0025** relative to gated CE. Its paired seed differences were +0.0002, -0.0038 and -0.0039, with magnitudes comparable to the seed SDs. Weights 0.3 and 1.0 gave means of 0.7224 and 0.7141. These validation results show no consistent gain from OP. Compact class clusters may obscure distinct targets, visual contexts and forms of irony within each label.

5. Six validation examples

Seed 42, gated CE+OP (weight 0.1), threshold 0.5. We inspected the two most confident false positives and false negatives, then two correct fusion decisions with the largest score shift from either unimodal model.

ID	Gold	CLIP	MPNet	FOP	Inspection
73021	0	0.708	0.826	0.990	A shrug accompanies a reference to Jews disappearing. The model may read an implied threat into the caption.
34975	0	0.762	0.872	0.980	A reaction photo frames a religious parody. The sexual-orientation reference may drive a false alarm.
76921	1	0.047	0.107	0.002	The caption excuses a threatening gesture toward a woman. Its ordinary wording can hide the implied justification.
82301	1	0.203	0.195	0.003	A howling dog accompanies an elongated racial slur. The altered spelling and benign scene may hide the cue.
19536	1	0.052	0.664	0.896	The caption and exaggerated nose invoke an antisemitic stereotype about money. Fusion corrects the image decision.
05349	0	0.889	0.501	0.058	The photo shows two women sitting by a wall. The country caption supplies no explicit insult. Fusion corrects both decisions.

These examples illustrate possible failure modes. They do not estimate their prevalence.

Group contributions

Assigned roles. Bogdan Fanea handled the experimental runs, including feature alignment, baseline reproduction, ablation training, metric verification, checkpoint checks and report formatting. Lena Bryll worked on the FOP architecture, loss and training code. Natalie Stadler worked on qualitative error analysis and report writing.

Sources and reused implementations

We reused Assignment 1 feature caches and baseline scores. The NumPy implementation adapts the official FOP fusion and loss code [2]. Checkpoints, seed histories and test predictions accompany the report.

AI usage

We used ChatGPT to help develop and debug the NumPy implementation and to write checks for feature alignment, checkpoint reproducibility and prediction outputs. We also used it to draft and revise explanations of the methods and results, and to adapt the report layout to match the Assignment 1 example. The reported scores and predictions were computed by running the assignment code.

References

[1] M. S. Saeed et al. Fusion and Orthogonal Projection for Improved Face-Voice Association. ICASSP, 2022, pp. 7057-7061. arXiv:2112.10483.

[2] Saeed et al. Official FOP implementation. Projection and fusion blocks in retrieval_model.py and OP loss in main.py.

[3] D. Kiela et al. The Hateful Memes Challenge: Detecting Hate Speech in Multimodal Memes. NeurIPS, 2020. arXiv:2005.04790.

[4] OpenAI. CLIP ViT-B/32 checkpoint. Revision 3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268. The frozen visual projection produces 512 features.

[5] Sentence Transformers. all-mpnet-base-v2 checkpoint. Revision e8c3b32edf5434bc2275fc9bab85f82640a19130.